Hackathon 6

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Our product

1. Python filtering

Filters relevant English transport and safety posts continuously from the live stream.

2. Hugging face model

Labels their sentiment (negative, neutral, positive).

cardiffnlp/twitter-roberta-base-sentiment-latest.

3. SQLite

Saves results, timestamps, sentiment scores, and stable post links

4. Streamlit

Displays live charts, trends over time, and raw posts interactively on the dashboard.

The problem:

- Passengers often feel unsafe at public transport stations at night due to poor lighting, isolation, and limited staff presence.
- Fear of harassment, intimidation, or unwanted encounters reduces passengers' sense of security while waiting for trains.
- Quiet or deserted stations at night can make passengers feel vulnerable, especially when travelling alone.

Target User

Public transport safety analyst at a rail operator reviewing online passenger feedback.

Core Decision

Inspect negative sentiment spikes, examine underlying posts, and decide whether concerns merit operational follow-up.

What is it NOT for?

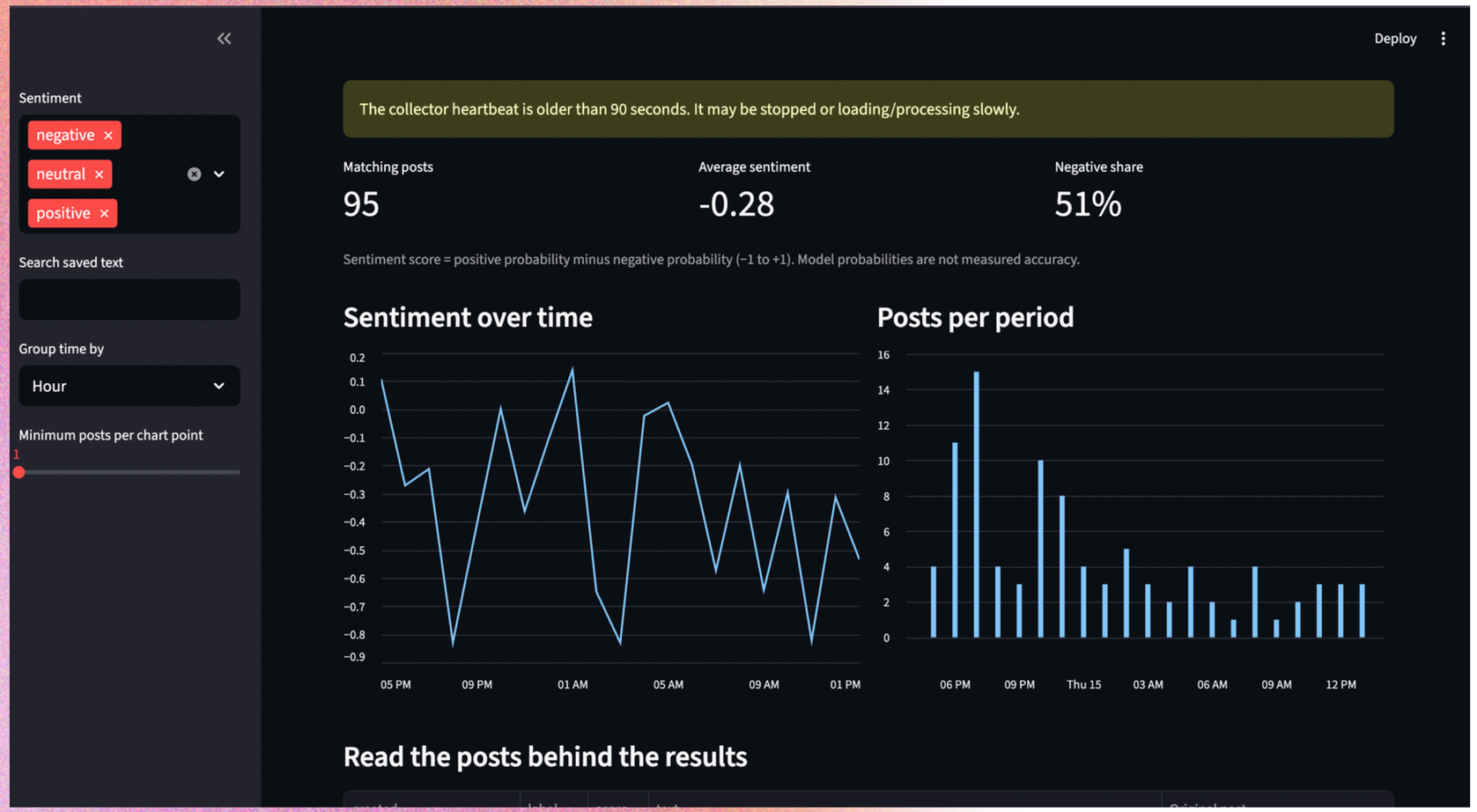
Not an emergency tool or automated safety ranking. Human verification required before action.

Why this problem/user?

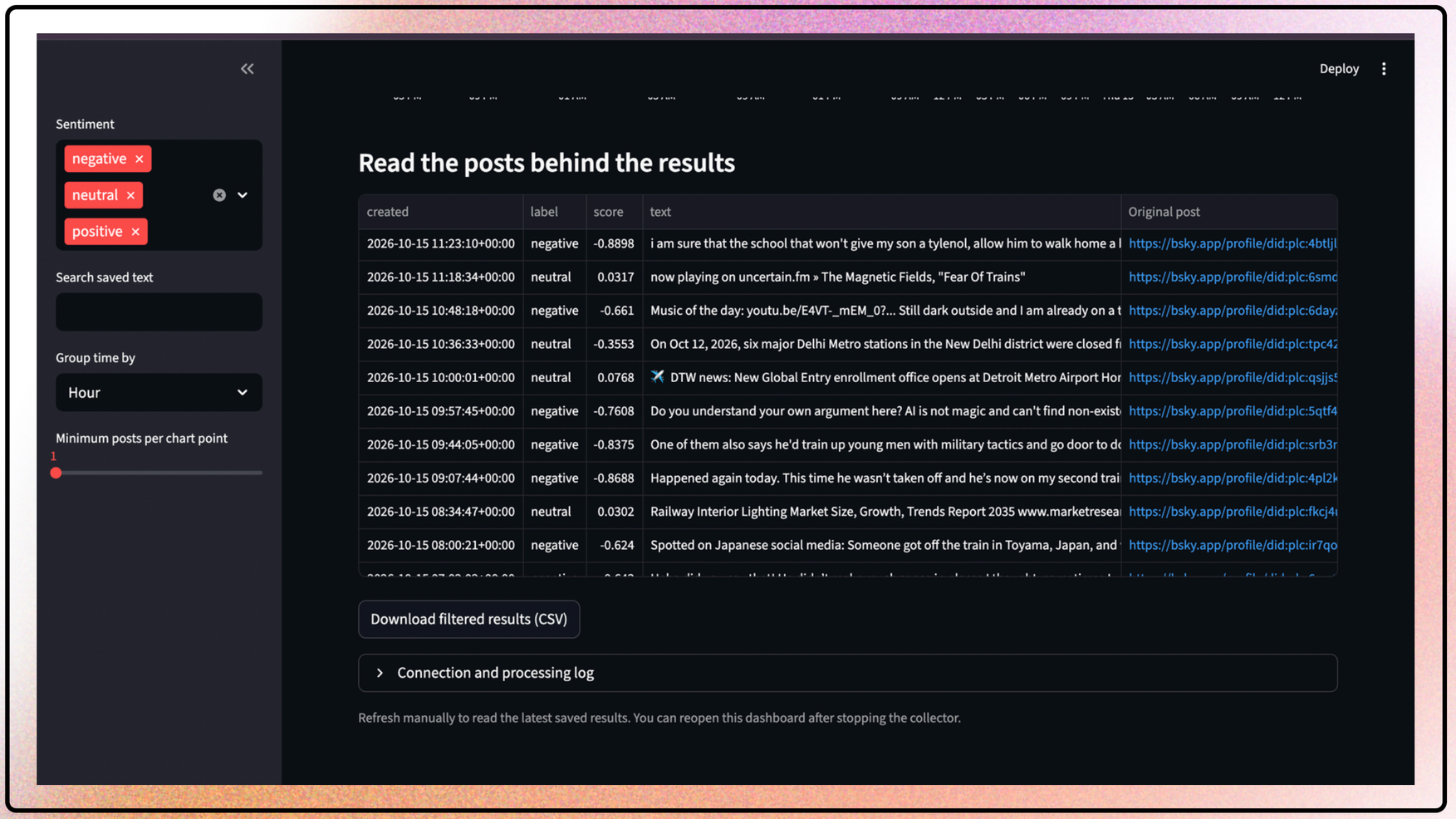
- Passenger safety: Concerns over darkness, isolation, and harassment undermine confidence in public transport.
- Impact: Unaddressed safety fears reduce transit adoption and restrict mobility.

Live dashboard + charts

http://localhost:8501/



Posts behind the results



Insights from our mistakes

1. Matching irrelevant posts

What happened: Keyword matching captured irrelevant posts like news headlines, song titles, or general metaphors.

Insight: Keywords alone are too broad for precise incident detection without context checks.

2. Misclasifying

What happened: Sentiment models misclassify slang, sarcasm, and complex transport complaints.

Insight: Generic models cannot be assumed accurate without testing against domain-specific data.

3. Human in-the-loop

What happened: Automated pipeline scores cannot automatically verify stations or justify security decisions.

Insight: Charts highlight where to look, but human review of source text is mandatory.

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Thank you for your attention

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